Technical MCQ - Pseudo Codes Based on PYQ 02 — Solutions & Explanations

🎯 Infosys Systems Engineer (SE) | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Total Questions: 10 | **Total Marks:** 13

Section: Technical Pseudo Codes -01

Problem 1 • What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print(“True”); Else Print(“False”); return 0; }

◯ PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print(“True”); Else Print(“False”); return 0; }

A True

C Error

D -1

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 2 • What is the output of following code: Int main(){ Int n=(1<<2+3<<4) Print(n); Return 0; }

◯ PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

What is the output of following code: Int main(){ Int n=(1<<2+3<<4) Print(n); Return 0; }

A 112

C 52

D 0

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 3 • What is the output of following: Integer x, y Set x = 15, y = 12 y = x – 1 do{ Print x x = y + (x – 2) } while(x < 40) end do while

◯ PRACTICE VARIANT

ID: Q3 Source: Infosys PYQ Level: Infosys SE Standard

What is the output of following: Integer x, y Set x = 15, y = 12 y = x – 1 do{ Print x x = y + (x – 2) } while(x < 40) end do while

A 14 26

C 27 39

D 15

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 4 • What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If End Function

◯ PRACTICE VARIANT

ID: Q4 Source: Infosys PYQ Level: Infosys SE Standard

What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If End Function

A 1 2 3 4 5

B 5 4 3 2 1

D -1

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 5 • int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }

◯ PRACTICE VARIANT

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }

A 0

C 5

D Compilation Error

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 6 • 1. Integer a = 5, b = 9 2. a = a ^ b 3. b = a ^ b 4. a = a ^ b 5. if (a == 9 && b == 5) 6. Print "Swapped" 7. else 8. Print "Not swapped" ◯ PRACTICE VARIANT

ID: Q6 Source: Infosys PYQ Level: Infosys SE Standard

1. Integer a = 5, b = 9 2. a = a ^ b 3. b = a ^ b 4. a = a ^ b 5. if (a == 9 && b == 5) 6. Print "Swapped" 7. else 8. Print "Not swapped"

A Line 2

B Line 3

C Line 4

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 7 • int main() { int x = 9, y = 2, z = 6; int a = x & y | z; printf("%d", a); return 0; } ◯ PRACTICE VARIANT

ID: Q7 Source: Infosys PYQ Level: Infosys SE Standard

int main() { int x = 9, y = 2, z = 6; int a = x & y | z; printf("%d", a); return 0; }

A Error

B 2

C 3

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 8 • Integer p, q, r Set p = 4, q = 2, r = 5 p = (q + p) + p if ((q + p) < (r - q) && 3 > p) then p = (q + r) + p if (((p + r) + (r + q)) <= (p + r + q)) then r = q + p else r = (p + r) + p end if p = (9 & 6) + q else q = r & r if ((q - 5) < (5 - q)) then r = (6 + 6) & q q = (q + 7) + r end if p = (1 ^ 11) + q end if Print p + q + r ◯ PRACTICE VARIANT

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

Integer p, q, r Set p = 4, q = 2, r = 5 p = (q + p) + p if ((q + p) < (r - q) && 3 > p) then p = (q + r) + p if (((p + r) + (r + q)) <= (p + r + q)) then r = q + p else r = (p + r) + p end if p = (9 & 6) + q else q = r & r if ((q - 5) < (5 - q)) then r = (6 + 6) & q q = (q + 7) + r end if p = (1 ^ 11) + q end if Print p + q + r

A undefined

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 9 • Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c > b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c < a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a Print a + b + c

◯ PRACTICE VARIANT

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

Integer a, b, c Set a = 2, b = 4, c = 7 b = 7 + a if (c > b) b = b + b end if b = 6 + a if ((a + b) < (b - a) || c < a) b = (c + 4) + c else a = c + b end if c = (c + 9) ^ a Print a + b + c

A undefined

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 10 • Integer a, b, c Set a = 1, b = 6, c = 7 b = (a & b) + b for (each c from 3 to 5) { b = (a + a) ^ a b = (12 + 5) + a } a = (a ^ 9) + a Print a + b

◯ PRACTICE VARIANT

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

Integer a, b, c Set a = 1, b = 6, c = 7 b = (a & b) + b for (each c from 3 to 5) { b = (a + a) ^ a b = (12 + 5) + a } a = (a ^ 9) + a Print a + b

A undefined

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.